

Allampati Bharath Kumar Reddy

bharath02032508@gmail.com | +91-8977984507 | [LinkedIn](#) | [GitHub](#)

ABOUT ME

Computer Science graduate with hands-on experience building and deploying full-stack applications using Java, JavaScript, React, Node.js, SQL, and cloud platforms. Strong foundation in object-oriented programming, data structures, databases, and software development fundamentals. Built production-style projects featuring authentication, REST-based integrations, AI capabilities, real-time features, and cloud deployment. Seeking a Software Engineering role focused on building scalable, reliable, and high-quality applications.

PROJECTS

Aether Cast — Podcast Streaming Platform ([link](#))

React.js, Node.js, Express.js, MongoDB Atlas, Supabase Auth, Gemini API

- Built a full-stack podcast platform with React 18, Node.js/Express, MongoDB Atlas, and Supabase authentication, featuring a global sticky audio player, PWA offline support, and live synced transcripts. Added automated API tests for authentication, validation, and core backend workflows.
- Integrated Google Gemini 1.5 Flash API for AI-powered episode summaries and smart tagging, along with real-time playback synchronization, creator drafts workspace, and in-browser recording with waveform visualization.
- Deployed frontend on Vercel and backend on Render with a full admin command center for platform-wide metrics, user management, and content moderation.

GATE Flow (In Progress) ([link](#))

React.js, JavaScript, CSS, Firebase, Gemini API, AI Integration

- Developed a comprehensive GATE CS & IT preparation platform featuring topic-wise practice, customizable CBT-style mock tests, performance analytics, and an AI-powered study assistant.
- Built configurable mock test workflows with subject selection, question navigation, negative marking, mistake analysis, and an integrated calculator aligned with GATE exam patterns.
- Integrated Firebase database and authentication systems to persist user test performance data and subject-wise analytics.

Digital Watermarking System ([link](#))

Python, Image Processing, LSB, DCT, QR Encoding, ML

- Implemented a robust digital watermarking framework supporting both LSB-based and DCT-based embedding techniques for securely hiding QR-encoded data inside color images.
- Designed end-to-end watermark workflows including QR generation, bit-level encoding, block-wise DCT coefficient manipulation, and image reconstruction with minimal perceptual distortion.
- Evaluated watermark quality using PSNR and tested extraction robustness under compression, noise, and format transformations.

SKILLS

- **Programming Languages:** Java, JavaScript, SQL, Python(basics)
- **Core Java:** OOP, Collections Framework, Exception Handling, Stream API
- **Frontend Technologies:** HTML5, CSS3, JavaScript, Bootstrap, React.js
- **Backend & Framework:** Spring Framework, Node.js, Express.js
- **Database:** MySQL, PostgreSQL, MongoDB
- **Currently Learning:** Spring Boot, RESTful API's, Multithreading
- **Relevant Coursework:** Data Structures & Algorithms, Database Management, Operating Systems
- **Version Control & Tools:** Git, GitHub

EDUCATION

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|---|-------------------------|
| • Vellore Institute of Technology | 2022 – June 2026 |
| B. Tech in Computer Science (Specialization: Information Security), 8.39/10 CGPA | Vellore, Tamil Nadu |
| • Narayana Junior College | 2020 – 2022 |
| Class 12(State Board), 92.2% Higher Secondary | Nellore, Andhra Pradesh |
| • Narayana High School | 2019 – 2020 |
| Class 10(State Board), 99.5 % | Nellore, Andhra Pradesh |

CERTIFICATES AND TRAINING

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|---|----------------------|
| • Oracle OCI Generative AI Professional -- Oracle 2025 Credential (link) | May 2025 – June 2025 |
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